Rates of Change & Concavity

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Average rate of change**

Find the average rate of change of $f(x) = x^2 + 2x$ on the interval $[1, 4]$.

AROC $= \frac{f(4) - f(1)}{4 - 1} = \frac{24 - 3}{3} = \frac{21}{3} = 7$.

2 **[Foundational]** • **Target: Average rate of change**

A population grows according to $P(t) = 200(1.05)^t$ (years). Find the average rate of change of the population from $t = 0$ to $t = 10$, to the nearest whole number.

$P(0) = 200$. $P(10) = 200(1.05)^{10} \approx 200(1.6289) \approx 325.8$. AROC $= \frac{325.8 - 200}{10} \approx 12.6$ people/year.

3 **[Foundational]** • **Target: Increasing/decreasing behavior from rate of change**

A function f has average rates of change of 2 on $[0, 1]$, 5 on $[1, 2]$, and 9 on $[2, 3]$. Determine whether f appears to be concave up or concave down on $[0, 3]$, and explain.

Since the average rate of change is itself increasing ($2 \to 5 \to 9$), f is speeding up as it increases — this is the signature of a concave-up function on $[0, 3]$.

4 **[Foundational]** • **Target: Increasing/decreasing behavior from rate of change**

A different function g has average rates of change of 8 on $[0, 1]$, 5 on $[1, 2]$, and 1 on $[2, 3]$. Classify the concavity of g on $[0, 3]$.

The average rate of change is decreasing ($8 \to 5 \to 1$), meaning g's growth is slowing down — this indicates g is concave down on $[0, 3]$.

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5 [Foundational] • Target: Visualizing concavity and inflection

The graph shows $f(x) = x^3 - 3x$, which changes concavity at $x = 0$.

a Describe the concavity of the graph on $(-2, 0)$. On $(-2, 0)$, the curve bends downward (concave down) — the graph lies below its tangent lines there.

b Describe the concavity of the graph on $(0, 2)$. On $(0, 2)$, the curve bends upward (concave up) — the graph lies above its tangent lines there.

c State what kind of point $x = 0$ represents. $x = 0$ is an inflection point: the location where concavity changes from down to up.

(a) On $(-2, 0)$, the curve bends downward (concave down) — the graph lies below its tangent lines there.; (b) On $(0, 2)$, the curve bends upward (concave up) — the graph lies above its tangent lines there.; (c) $x = 0$ is an inflection point: the location where concavity changes from down to up.

6 [Foundational] • Target: Rate of change of the rate of change

Using average rates of change over consecutive equal-width intervals as a proxy for how the function's growth rate itself is changing, explain what it means for a function to be "concave up" purely in terms of consecutive AROC values, without any calculus notation.

A function is concave up on an interval when its average rate of change is itself increasing as you move from left to right across consecutive equal-width sub-intervals — the function isn't just increasing, it's increasing at a faster and faster pace (or decreasing at a slower and slower pace).

7 [Foundational] • Target: Applying rate-of-change reasoning to models

A company's revenue (in thousands) is modeled by $R(t) = t^3 - 6t^2 + 40t$ for $0 \le t \le 10$ years. Find the average rate of change on $[0, 2]$ and on $[2, 4]$, then determine whether revenue growth is accelerating or decelerating over $[0, 4]$.

$R(0) = 0$, $R(2) = 8 - 24 + 80 = 64$, $R(4) = 64 - 96 + 160 = 128$. AROC on $[0, 2]$: $\frac{64 - 0}{2} = 32$. AROC on $[2, 4]$: $\frac{128 - 64}{2} = 32$. Since both average rates are equal, revenue growth appears roughly linear (neither clearly accelerating nor decelerating) over this specific interval pair.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Applying rate-of-change reasoning to models

For the same revenue function, find the AROC on $[4, 6]$ and compare it to the AROC on $[2, 4]$ found above, to determine the concavity trend as t increases past 4.

$R(6) = 216 - 216 + 240 = 240$. AROC on $[4, 6]$: $\frac{240 - 128}{2} = 56$. Since $56 > 32$ (the AROC on $[2, 4]$), the rate of change is increasing, indicating the revenue curve is concave up on $[4, 6]$ — growth is accelerating in this later interval.

9 [Intermediate] • Target: Estimating rate of change from a table

A car's odometer reading (miles) is recorded every hour: $t = 0$: 120; $t = 1$: 175; $t = 2$: 238; $t = 3$: 310. Find the AROC over each hour and determine whether the car's speed is increasing, decreasing, or constant.

AROC on $[0, 1]$: $175 - 120 = 55$ mph. AROC on $[1, 2]$: $238 - 175 = 63$ mph. AROC on $[2, 3]$: $310 - 238 = 72$ mph. Since the AROC is increasing each hour ($55 \to 63 \to 72$), the car's speed is increasing — consistent with a concave-up distance function.

10 [Intermediate] • Target: Estimating rate of change from a table

Using the same table, find the AROC over the full interval $[0, 3]$ and explain why it differs from the individual hourly rates.

AROC on $[0, 3]$: $\frac{310 - 120}{3} = \frac{190}{3} \approx 63.3$ mph. This is an average over the whole trip, blending the slower early hour with the faster later hour — it necessarily falls between the smallest (55) and largest (72) hourly rates, but doesn't equal any single hourly rate.

11 [Intermediate] • Target: Visualizing rate of change on a graph

The graph shows $f(x) = \sqrt{x}$, whose average rate of change decreases as x increases.

a Find the AROC of $f(x) = \sqrt{x}$ on $[0, 1]$ and on $[4, 9]$. AROC on $[0, 1]$: $\frac{\sqrt{1} - \sqrt{0}}{1} = 1$. AROC on $[4, 9]$: $\frac{\sqrt{9} - \sqrt{4}}{5} = \frac{3 - 2}{5} = 0.2$.

b Explain how the shrinking steepness in the graph relates to your two AROC values. The curve rises steeply near $x = 0$ and flattens out as x increases, matching the drop from an AROC of 1 down to 0.2 — the function is growing, but at an ever-slower pace.

c State whether $f(x) = \sqrt{x}$ is concave up or concave down, based on this pattern. Since the AROC is decreasing as x increases, $f(x) = \sqrt{x}$ is concave down on $(0, \infty)$.

(a) AROC on $[0, 1]$: $\frac{\sqrt{1} - \sqrt{0}}{1} = 1$. AROC on $[4, 9]$: $\frac{\sqrt{9} - \sqrt{4}}{5} = \frac{3 - 2}{5} = 0.2$.; (b) The curve rises steeply near $x = 0$ and flattens out as x increases, matching the drop from an AROC of 1 down to 0.2 — the function is growing, but at an ever-slower pace.; (c) Since the AROC is decreasing as x increases, $f(x) = \sqrt{x}$ is concave down on $(0, \infty)$.

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12 **[Intermediate]** • **Target: Concavity and real-world interpretation**

A company reports that its costs are increasing, but at a decreasing rate. Sketch what this implies about the concavity of the cost function, and explain your reasoning without using calculus notation.

"Increasing but at a decreasing rate" means the function is going up (each AROC is positive) but each successive AROC is smaller than the one before — this is the definition of concave down. So the cost function's graph rises but bends downward, flattening out over time.

13 **[Intermediate]** • **Target: Average rate of change**

Find the average rate of change of the polar radius $r(\theta) = 3 + 2\sin(2\theta)$ over the interval $[0, \frac{\pi}{4}]$, and interpret whether the distance from the origin is increasing or decreasing on this interval.

Step 1: Evaluate $r(\theta)$ at $\theta = 0$ and $\theta = \pi/4$: $r(0) = 3 + 2\sin(0) = 3, \quad r(\frac{\pi}{4}) = 3 + 2\sin(\frac{\pi}{2}) = 3 + 2 = 5.$ Step 2: Compute the average rate of change: $\frac{r(\pi/4) - r(0)}{\pi/4 - 0} = \frac{5-3}{\pi/4} = \frac{8}{\pi} \approx \mathbf{2.546}$ units/rad. Because $r(\theta)$ is strictly increasing and positive on $[0, \pi/4]$, the curve moves further away from the pole.

14 **[Intermediate]** • **Target: Increasing/decreasing behavior from rate of change**

Solve the exponential equation $9^x - 4 \cdot 3^{x+1} + 27 = 0$ for all real x.

Step 1: Rewrite in terms of $u = 3^x$: Note that $9^x = (3^x)^2 = u^2$ and $3^{x+1} = 3 \cdot 3^x = 3u$: $u^2 - 12u + 27 = 0.$ Step 2: Factor the quadratic: $(u-3)(u-9) = 0 \Rightarrow u = 3 \quad \text{or} \quad u = 9.$ Step 3: Back-substitute $3^x = u$: $3^x = 3^1 \Rightarrow x = 1; \quad 3^x = 3^2 \Rightarrow x = 2.$ Solutions: $\mathbf{x = 1, 2}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Visualizing concavity and inflection** • **Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions**

Evaluate $\log_2(5) \cdot \log_5(9) \cdot \log_3(16)$ without a calculator using the change-of-base formula.

Step 1: Convert all logarithms to natural logarithms: $\frac{\ln 5}{\ln 2} \cdot \frac{\ln 9}{\ln 5} \cdot \frac{\ln 16}{\ln 3}.$ Step 2: Simplify using prime power identities $\ln 9 = 2\ln 3$ and $\ln 16 = 4\ln 2$: $\frac{\ln 5}{\ln 2} \cdot \frac{2\ln 3}{\ln 5} \cdot \frac{4\ln 2}{\ln 3} = 2 \times 4 = \mathbf{8}$.

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16 [Challenging] • Target: Rate of change of the rate of change • Cross-Lesson: 4.03 Parametric Equations & Modeling

A right-angled triangle ABC is shown with the right angle at vertex C. The length of side AC is 8 units and the length of side BC is 6 units. What is the value of $\sin(\angle BAC)$?

The triangle ABC is a right-angled triangle with the right angle at C. We are given AC = 8 and BC = 6. We want to find $\sin(\angle BAC)$. In a right-angled triangle, the sine of an angle is defined as the ratio of the length of the side opposite the angle to the length of the hypotenuse. First, we need to find the length of the hypotenuse AB using the Pythagorean theorem: $AB^2 = AC^2 + BC^2$. $AB^2 = 8^2 + 6^2$ $AB^2 = 64 + 36$ $AB^2 = 100$ $AB = \sqrt{100} = 10$ units. Now, we can find $\sin(\angle BAC)$: $\sin(\angle BAC) = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{BC}{AB}$ $\sin(\angle BAC) = \frac{6}{10} = \frac{3}{5}$. Therefore, the value of $\sin(\angle BAC)$ is $\frac{3}{5}$.

17 [Challenging] • Target: Applying rate-of-change reasoning to models • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A point P lies on the unit circle such that its terminal ray makes an angle of θ with the positive x-axis. If the x-coordinate of P is $\frac{1}{2}$, what is the y-coordinate of P? Assume θ is in the first quadrant.

For a point P on the unit circle, its coordinates are given by $(\cos\theta, \sin\theta)$. We are given that the x-coordinate of P is $\frac{1}{2}$, so $\cos\theta = \frac{1}{2}$. Since P is on the unit circle, its coordinates satisfy the equation $x^2 + y^2 = 1$. Substituting the x-coordinate, we get $(\frac{1}{2})^2 + y^2 = 1$. This simplifies to $\frac{1}{4} + y^2 = 1$. Subtracting $\frac{1}{4}$ from both sides gives $y^2 = 1 - \frac{1}{4} = \frac{3}{4}$. Taking the square root of both sides, we get $y = \pm\frac{\sqrt{3}}{2}$. Since the problem states that θ is in the first quadrant, the y-coordinate must be positive. Therefore, the y-coordinate of P is $\frac{\sqrt{3}}{2}$.

18 [Challenging] • Target: Estimating rate of change from a table • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Find the exact value of $\cos\left(\frac{7\pi}{12}\right)$ and $\sin\left(\frac{7\pi}{12}\right)$ using angle addition identities.

Step 1: Decompose $\frac{7\pi}{12} = \frac{\pi}{3} + \frac{\pi}{4}$: Step 2: Apply cosine and sine sum formulas:
$\cos\left(\frac{7\pi}{12}\right) = \cos\frac{\pi}{3}\cos\frac{\pi}{4} - \sin\frac{\pi}{3}\sin\frac{\pi}{4} = \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) - \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{\mathbf{2}} - \sqrt{\mathbf{6}}}{\mathbf{4}}$.
$\sin\left(\frac{7\pi}{12}\right) = \sin\frac{\pi}{3}\cos\frac{\pi}{4} + \cos\frac{\pi}{3}\sin\frac{\pi}{4} = \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) + \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{\mathbf{6}} + \sqrt{\mathbf{2}}}{\mathbf{4}}$.

19 [Challenging] • Target: Visualizing rate of change on a graph • Cross-Lesson: 2.01 Exponential Functions & Growth Models

A Ferris wheel has a diameter of 40 meters and its lowest point is 4 meters above the ground. Passengers board at the lowest point at $t = 0$, and the wheel completes one full revolution every 8 minutes. Write a cosine model $h(t)$ for a passenger's height at time t, and find the times during the first revolution when $h(t) = 34$ meters.

Step 1: Determine sinusoidal parameters: Radius (amplitude) $= 20$ m; midline $D = 4 + 20 = 24$ m; angular frequency $B = \frac{2\pi}{8} = \frac{\pi}{4}$. Starting at the minimum gives a negative cosine:
$\mathbf{h(t) = -20\cos\left(\frac{\pi}{4}t\right) + 24}$. Step 2: Solve $h(t) = 34$ on $[0, 8]$:
$-20\cos\left(\frac{\pi}{4}t\right) + 24 = 34 \Rightarrow \cos\left(\frac{\pi}{4}t\right) = -\frac{1}{2}$. $\frac{\pi}{4}t = \frac{2\pi}{3}, \frac{4\pi}{3} \Rightarrow \mathbf{t = \frac{8}{3}}$ min and $\mathbf{t = \frac{16}{3}}$ min.

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20 [Challenging] • Target: Concavity and real-world interpretation • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Find the exact angle θ between $\mathbf{u} = \langle 3, 1\rangle$ and $\mathbf{v} = \langle 1, 2\rangle$, and compute the vector projection $\text{proj}_{\mathbf{v}}\mathbf{u}$.

Step 1: Compute dot product and magnitudes: $\mathbf{u} \cdot \mathbf{v} = 3(1) + 1(2) = 5, \quad ||\mathbf{u}|| = \sqrt{10}, \quad ||\mathbf{v}|| = \sqrt{5}.$ Step 2: Angle between vectors: $\cos\theta = \frac{5}{\sqrt{10}\sqrt{5}} = \frac{5}{5\sqrt{2}} = \frac{\sqrt{2}}{2} \Rightarrow \mathbf{\theta = \frac{\pi}{4} = 45^\circ}$. Step 3: Vector projection onto $\mathbf{v}$: $\text{proj}_{\mathbf{v}}\mathbf{u} = \left(\frac{\mathbf{u}\cdot\mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v} = \frac{5}{5}\langle 1, 2\rangle = \mathbf{\langle 1, 2\rangle}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Average rate of change

Find the average rate of change of $f(x) = -7x^2$ on $[-4, -1]$.

a. 35 ✓ **b.** -21

c. 33 **d.** 37

$\text{AROC} = \frac{f(-1) - f(-4)}{-1 - (-4)} = \frac{-7 - (-112)}{3} = 35.$

22 [Foundational] • Target: Increasing/decreasing behavior from rate of change

Find the average rate of change of $f(x) = 9x^2 + 2x$ on $[5, -3]$.

a. 18 **b.** 20 ✓

c. -70 **d.** 22

$\text{AROC} = \frac{f(-3) - f(5)}{-3 - (5)} = \frac{75 - (235)}{-8} = 20.$

23 [Foundational] • Target: Visualizing concavity and inflection

Find the average rate of change of $f(x) = -9x^2 + x$ on $[4, 0]$.

a. -35 ✓ **b.** -33

c. -37 **d.** 37

$\text{AROC} = \frac{f(0) - f(4)}{0 - (4)} = \frac{0 - (-140)}{-4} = -35.$

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24 [Foundational] • Target: Rate of change of the rate of change

Find the average rate of change of $f(x) = 2x^2 - 5x$ on $[6, -3]$.

a. -1

b. -23

c. 1 ✓

d. 3

AROC $= \frac{f(-3) - f(6)}{-3 - (6)} = \frac{33 - (42)}{-9} = 1.$

25 [Foundational] • Target: Applying rate-of-change reasoning to models

Find the average rate of change of $f(x) = 6x^2$ on $[3, -3]$.

a. 0 ✓

b. -36

c. 2

d. -2

AROC $= \frac{f(-3) - f(3)}{-3 - (3)} = \frac{54 - (54)}{-6} = 0.$

26 [Foundational] • Target: Estimating rate of change from a table

Find the average rate of change of $f(x) = -9x^2 - 9x$ on $[-4, 0]$.

a. -45

b. 25

c. 29

d. 27 ✓

AROC $= \frac{f(0) - f(-4)}{0 - (-4)} = \frac{0 - (-108)}{4} = 27.$

27 [Foundational] • Target: Visualizing rate of change on a graph

Find the average rate of change of $f(x) = x^2 + 4x$ on $[5, 4]$.

a. 11

b. 15

c. 13 ✓

d. 3

AROC $= \frac{f(4) - f(5)}{4 - (5)} = \frac{32 - (45)}{-1} = 13.$

28 [Foundational] • Target: Concavity and real-world interpretation

Find the average rate of change of $f(x) = -2x^3$ on $[2, -5]$.

a. -38 ✓

b. -42

c. -18

d. -34

AROC $= \frac{f(-5) - f(2)}{-5 - (2)} = \frac{250 - (-16)}{-7} = -38.$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Average Rate of Change (AROC)**

Find the average rate of change of $f(x) = -x^3$ on $[0, 5]$.

a. -21

b. -25 ✓

c. -27

d. -23

AROC $= \frac{f(5) - f(0)}{5 - (0)} = \frac{-125 - (0)}{5} = -25.$

30 **[Intermediate]** • **Target: Increasing/Decreasing and Changing Rates**

Find the average rate of change of $f(x) = 3x^3$ on $[-2, 3]$.

a. 27

b. 15

c. 21 ✓

d. 3

AROC $= \frac{f(3) - f(-2)}{3 - (-2)} = \frac{81 - (-24)}{5} = 21.$

31 **[Intermediate]** • **Target: Concavity & Inflection Points**

Find the average rate of change of $f(x) = 2x^3$ on $[-4, 1]$.

a. 30

b. 22

c. 26 ✓

d. 18

AROC $= \frac{f(1) - f(-4)}{1 - (-4)} = \frac{2 - (-128)}{5} = 26.$

32 **[Intermediate]** • **Target: Contextual Rates of Change Applications**

Find the average rate of change of $f(x) = -5x^3$ on $[-2, 3]$.

a. -25

b. -5

c. -35 ✓

d. -45

AROC $= \frac{f(3) - f(-2)}{3 - (-2)} = \frac{-135 - (40)}{5} = -35.$

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33 [Intermediate] • Target: Average rate of change

Find the average rate of change of $f(x) = -5x^3$ on $[0, -3]$.

a. -47

b. -35

c. -55

d. -45 ✓

AROC $= \frac{f(-3) - f(0)}{-3 - (0)} = \frac{135 - (0)}{-3} = -45$.

34 [Intermediate] • Target: Increasing/decreasing behavior from rate of change

Find the average rate of change of $f(x) = 6x^3$ on $[3, -5]$.

a. 24

b. 102

c. 126

d. 114 ✓

AROC $= \frac{f(-5) - f(3)}{-5 - (3)} = \frac{-750 - (162)}{-8} = 114$.

35 [Intermediate] • Target: Visualizing concavity and inflection

A car's odometer reading (miles) is recorded every hour: $t = 0$: 75; $t = 1$: 111; $t = 2$: 138; $t = 3$: 156. Is the car's speed increasing, decreasing, or constant?

a. **Decreasing** ✓

b. Increasing

c. Constant

d. Cannot be determined from this data

AROC on $[0, 1]$: 36 mph. AROC on $[1, 2]$: 27 mph. AROC on $[2, 3]$: 18 mph. The hourly rates are decreasing, so the car's speed is Decreasing.

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36 [Intermediate] • Target: Rate of change of the rate of change

A car's odometer reading (miles) is recorded every hour: $t = 0$: 151; $t = 1$: 174; $t = 2$: 197; $t = 3$: 220. Is the car's speed increasing, decreasing, or constant?

a. Increasing

b. **Constant ✓**

c. Decreasing

d. Cannot be determined from this data

AROC on $[0, 1]$: 23 mph. AROC on $[1, 2]$: 23 mph. AROC on $[2, 3]$: 23 mph. The hourly rates are constant, so the car's speed is Constant.

37 [Intermediate] • Target: Applying rate-of-change reasoning to models

A function h has average rates of change of 4 on $[0, 1]$, 7 on $[1, 2]$, and 12 on $[2, 3]$. Classify the concavity of h on $[0, 3]$.

a. Linear (neither concave up nor down)

b. Cannot be determined from this data

c. Concave down

d. **Concave up ✓**

The average rate of change is increasing ($4 \to 7 \to 12$), so h is concave up on $[0, 3]$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Estimating rate of change from a table • Cross-Lesson: 1.02 Rational Functions & Asymptotes

A function h has average rates of change of 9 on $[0, 1]$, 8 on $[1, 2]$, and 6 on $[2, 3]$. Classify the concavity of h on $[0, 3]$.

a. Cannot be determined from this data

b. Linear (neither concave up nor down)

c. **Concave down ✓**

d. Concave up

The average rate of change is decreasing ($9 \to 8 \to 6$), so h is concave down on $[0, 3]$.

39 [Challenging] • Target: Visualizing rate of change on a graph • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Find the average rate of change of $f(x) = 9\sqrt{x}$ on $[1, 9]$.

a. $\frac{13}{4}$

b. $\frac{9}{4}$ ✓

c. $\frac{9}{2}$

d. $\frac{1}{4}$

AROC $= \frac{9\sqrt{9} - 9\sqrt{1}}{9 - 1} = \frac{27 - 9}{8} = \frac{9}{4}$.

40 [Challenging] • Target: Concavity and real-world interpretation • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Find the average rate of change of $f(x) = 5\sqrt{x}$ on $[16, 49]$.

a. $\frac{5}{11}$ ✓

b. $\frac{1}{11}$

c. $\frac{16}{11}$

d. $\frac{5}{3}$

AROC $= \frac{5\sqrt{49} - 5\sqrt{16}}{49 - 16} = \frac{35 - 20}{33} = \frac{5}{11}$.

41 [Challenging] • Target: Average Rate of Change (AROC) • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Calculate the average rate of change of $f(x) = 2x^2 - 3x + 1$ on the interval $[1, 4]$.

a. 7 ✓

b. 9

c. 21

d. 6

AROC $= \frac{f(4) - f(1)}{4 - 1}$. $f(4) = 2(16) - 12 + 1 = 21$. $f(1) = 2 - 3 + 1 = 0$. AROC $= \frac{21 - 0}{3} = 7$.

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42 [Challenging] • Target: Average Rate of Change (AROC) • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

For a linear function $g(x) = mx + b$, what can be said about the average rate of change on any interval $[a, b]$?

a. It is constant and always equal to m ✓

b. It depends on the midpoint $\frac{a+b}{2}$

c. It is zero if $b > 0$

d. It equals b

Linear functions have a constant rate of change equal to the slope m over any interval.

43 [Challenging] • Target: Increasing/Decreasing and Changing Rates • Cross-Lesson: 4.01 Inverse & Composite Functions

If $h(x)$ is decreasing at an increasing rate on an interval, what happens to the slope of secant lines?

a. The slopes become less negative (closer to zero) ✓

b. The slopes become more negative (steeper downward)

c. The slopes remain constant

d. The slopes switch from positive to negative

Decreasing means rates are negative. At an increasing rate means the rate values are increasing (e.g. from -10 to -4 to -1), making the slope less negative and flattening out (concave up).

44 [Challenging] • Target: Concavity & Inflection Points • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

On the interval $(-\infty, 0)$, f is concave up; on $(0, \infty)$, f is concave down. If $f(0) = 4$, which statement is true?

a. $(0, 4)$ is an inflection point of f ✓

b. $(0, 4)$ is a local minimum of f

c. $(0, 4)$ is a local maximum of f

d. $f'(0)$ must equal 0

Because the concavity changes at $x = 0$ and $(0, 4)$ lies on the continuous curve, $(0, 4)$ is an inflection point.

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45 [Challenging] • Target: Contextual Rates of Change Applications • Cross-Lesson: 4.03 Parametric Equations & Modeling

Water is poured into a conical cup at a constant rate. What is the concavity of the depth of water $h(t)$ as a function of time?

a. Concave down, because the radius increases with height so height increases at a decreasing rate ✓

b. Concave up, because height increases at an increasing rate

c. Linear, because the flow rate is constant

d. Constant concavity of zero

Because the cup widens as it fills, each additional volume unit adds progressively less height. Thus the rate of change of height decreases over time, giving a concave down graph.

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